Identification of an unknown sample through X-ray Fluorescence and Diffraction

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Abstract

A powdered unknown sample is investigated through X-Ray Fluorescence and Diffraction. Fluorescence is conducted with a radioactive ^{133}Ba source, for this reason the resulted spectrum is noisy; however suffieciently bright for our investigation. After a calibration of the channels of the amplifier via fluorescence of known sample, the material is examinated. A clear peak at 8.67 keV emerges, near the 8.64 keV Kα_1 peak of Zn. Diffraction results confirm the presence of Zn, reporting with extreme accuracy that the sample contains primarly ZnO.

Introduction

In an X-Ray Fluorescence (XRF) experiment a material is bombarded with high energy radiation, capable of extracting an electron from the first atomic bound state. The emission from the sample is peaked in the characteristic transitions of the outer shell electrons that fill the unoccupied state. Following the Siegbahn notation, the brightest peaks correspond to Kα transitions. The depth of the ionic potential grows with atomic number Z, therefore every element has a characteristic X-Ray spectrum.

In an X-Ray Diffraction (XRD) experiment the incident radiation interacts with the crystalline powdered sample, producing an interference pattern according to Bragg's Law. The acquired spectrum presents peaks at various angles and can be compared with a database of spectrum of known materials. XRF tells only which element is found inside the sample, while XRD gives information about lattice spacing, crystalline structure and type of bonding.

Materials and design

X-Ray photons of approximately 30 keV are generated by a radioactive ^{133}Ba source. The powdered sample is placed between the source and the Si detector: it absorbs a fraction of the incoming photons and re-emits at lower energies according to the tabulated energy spectrum [2] [3]. The experiment is not performed in vacuum, therefore low energy O and N K$_\alpha$ peaks are always present, because of the air. For this reason, in order to avoid saturating the first channels of the ADC, a low energy cut-off is performed. This apparatus therefore cuts also low energy lines that come from the sample and cannot tell if there is, for instance, any O in the sample. A channel-to-energy calibration is performed analyzing the spectrum of five known materials: Fe, Ge, Cu, Ag and Cs from the Ba radioactive source.

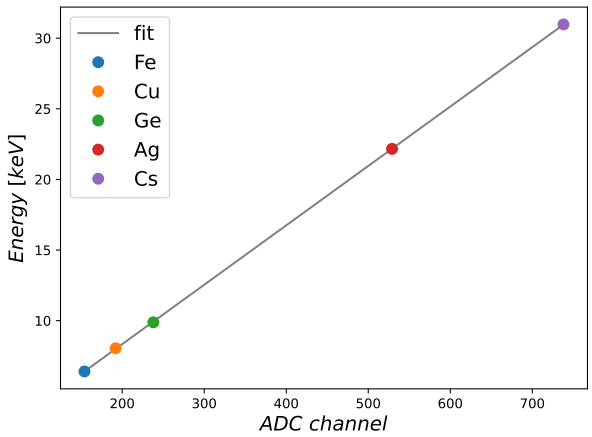


Figure 1: Calibration line obtained by fitting. Every point corresponds to the Kα_1 transition of the sample in the legend on the y-axis, with the corresponding ADC channel in which highest count is registered on the x-axis.

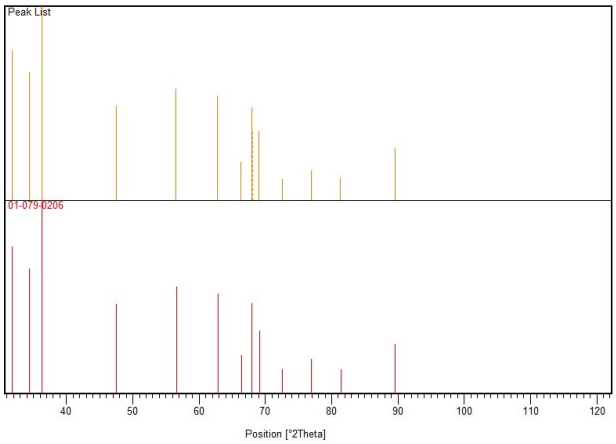


Figure 2: Comparison between peak list of our sample (upper) and ZnO (lower) [1].

Results and Discussion

Because of the weakness of the radioactive source, counts are collected in a period of time of 10 minutes. With reference to Fig. (4), in the XRF of the unknown material, a bright peak at about 8.67 keV is found, probably due to the Kα_1 transition of the unknown sample. The besst guess is the 8.64 keV [2] [3] Kα_1 transition line of Zn. The other elements with the nearest Kα_1 line are Cu (8.05 keV) and Ga (9.25 keV). For the reasons already explained, we cannot conclude that a lighter element, such as N or O, is present. In order to acquire informations about the crystalline structure, the lattice constant and the exact composition of the sample: diffraction peaks generate a unique fingerprint of the sample, that can be compared with a large database of samples, provided by softwares such as X'Pert HighScore. The software finds the most similar spectrum, given the angle position and amplitude of the peaks on the XRD spectrum. Fig. (2) shows that ZnO is the best guess for the unknown material.

Conclusions

Although the intensity of the radioactive source was very low compared with other setups, calibration peaks are sufficiently bright and Zn peaks is quite distinguished from the noise. Our XRF setup not only makes lighter elements undetectable, but also it is not possibe to exclude that the sample contains other heavier elements with atomic mass near Cesium; moreover, channels over 800 have been cut off because of a malfunctioning of the setup. XRD, however, furnishes more reliable results, consistent with XRF data.

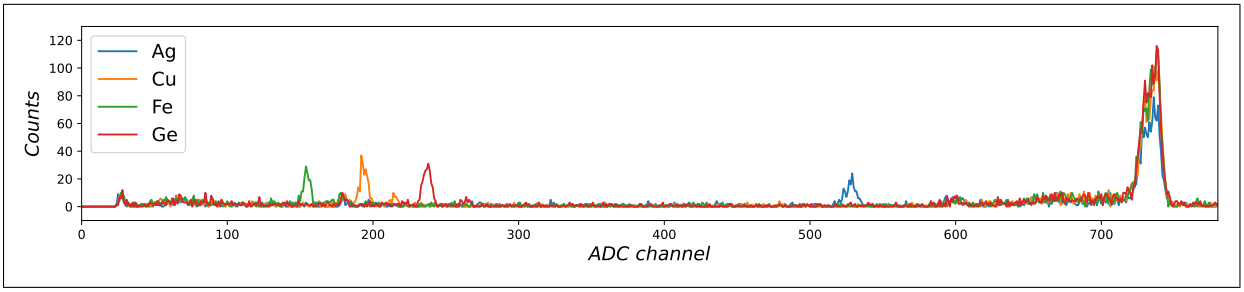


Figure 3: XRF spectrum of known samples. The most intense pick on the right is due to Ba source, the other four peaks with different colors are due to to the K_α transition of the corresponding element.

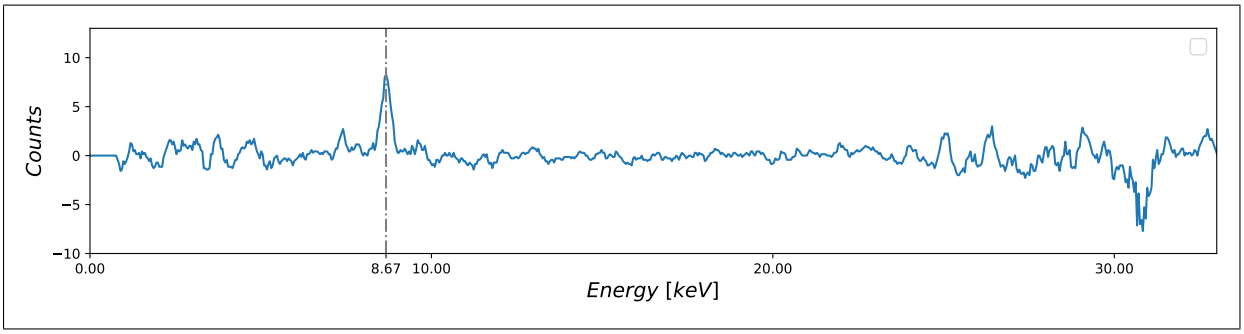


Figure 4: XRF spectrum of the unknown sample. The plot is obtained firstly by performing a moving average on both the spectrum of the unknown sample and the source; and then subtracting the counts of the source from the XRF of the sample. A sharp positive peak is found at 8.67 eV, as highlighted in the plot. The fluctuations in the right are due to the noise near the Cs $K\alpha_1$ line.

References

[1] J. Albertsson, S. C. Abrahams, and Å. Kvick. "Atomic displacement, anharmonic thermal vibration, expansivity and pyroelectric coefficient thermal dependences in ZnO". In: *Acta Crystallographica Section B* 45.1 (1989), pp. 34–40.

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[3] M. O. Krause and J. H. Oliver. "Natural widths of atomic K and L levels, Kα X-ray lines and several KLL Auger lines". In: *Journal of Physical and Chemical Reference Data* 8 (1979).